

SAULT STE MARIE SW PIER, MI, USA

WMO: 997268

Lat: 46.501N

Lon: 84.372W

Elev: 609

StdP: 14.38

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.1	(c) -4.2	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 47.5	(k) 16.7	(l) 36.4	(m) 19.1	(n) 6.6	(o) 100	(p) 0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.5	(c) 82.0	(d) N/A	(e) 79.1	(f) N/A	(g) 76.5	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 9.1	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 31.4	(b) 25.0	(c) 20.9	(e) DB	(f) -14.8	(g) 85.9	(h) 8.1	(i) 2.2	(j) -20.6	(k) 87.5	(l) -25.4	(m) 88.8	(n) -29.9	(o) 90.1	(p) -35.8	(q) 91.7	(r) N/A
(4) 31.4	(5) 25.0	(6) 20.9	(7) DB	(8) -14.8	(9) 85.9	(10) 8.1	(11) 2.2	(12) -20.6	(13) 87.5	(14) -25.4	(15) 88.8	(16) -29.9	(17) 90.1	(18) -35.8	(19) 91.7	(20) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.5	15.6	17.2	26.2	39.9	51.3	60.5	66.2	65.6	58.6	47.5	35.4	24.4
	DBStd	19.61	11.63	10.18	10.76	7.50	7.70	6.31	5.64	4.79	6.68	7.43	8.58	9.56
	HDD50	4504	1066	920	741	313	76	4	0	0	7	139	443	795
	HDD65	8427	1531	1340	1204	753	431	158	54	51	212	544	889	1260
	CDD50	1761	0	0	2	11	115	320	502	482	264	62	3	0
	CDD65	210	0	0	0	0	6	24	91	68	19	2	0	0
	CDH74	1109	0	0	0	0	69	197	503	271	64	5	0	0
	CDH80	166	0	0	0	0	12	32	84	32	6	0	0	0
(14) Wind	WSAvg	8.8	9.3	9.4	9.4	9.8	9.7	8.2	7.4	7.6	7.0	8.6	9.0	9.9
(15) Precipitation	PrecAvg	34.0	2.3	1.6	2.0	2.4	2.7	2.9	2.9	3.4	3.7	3.7	3.5	2.8
	PrecMax	48.8	4.5	3.7	5.0	5.5	7.4	7.4	7.2	9.5	8.3	7.6	7.7	6.3
	PrecMin	22.4	0.5	0.2	0.2	0.9	0.6	0.4	0.6	1.0	0.5	0.2	0.9	0.6
	PrecStd	5.6	0.9	0.8	1.1	1.0	1.3	1.4	1.4	1.5	1.8	1.7	1.3	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.1	40.9	59.9	67.4	81.7	84.4	86.6	83.9	80.2	73.3	61.5	45.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	38.2	36.1	50.9	61.4	74.0	78.7	82.5	80.2	75.4	67.4	52.8	41.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	35.7	33.7	45.5	56.2	68.6	75.8	79.6	77.3	72.2	62.3	49.3	38.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	33.0	31.7	40.0	51.6	63.9	72.2	76.8	74.4	69.2	58.3	46.5	36.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	MDBR		13.8	14.1	15.6	14.5	17.3	17.0	15.5	13.6	13.8	10.7	9.9	10.6
	5% DB	MCDBR	10.7	13.0	18.0	23.2	24.8	23.2	20.2	18.1	17.1	16.4	13.3	10.9
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.279	0.290	0.320	0.357	0.370	0.388	0.390	0.379	0.365	0.331	0.302	0.287
	taud		2.333	2.288	2.322	2.327	2.372	2.344	2.358	2.385	2.422	2.503	2.505	2.359
	Ebn at Noon		259	280	287	285	284	278	276	275	269	264	252	242
	Edh at Noon		23	30	34	37	37	38	37	35	31	25	20	21
(44) All-Sky Solar Radiation	RadAvg		345	631	1038	1413	1761	1897	1904	1609	1205	703	370	269
	RadStd		37	70	116	159	145	131	108	106	92	80	47	24

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page